



Fig. 6: Chaser Designer output window

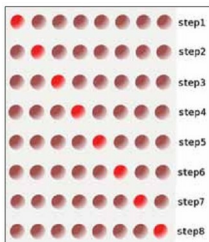


Fig. 7: Forward chasing effect pattern

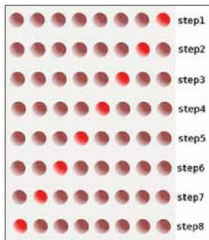


Fig. 8: Reverse LED pattern



Fig. 9: Dancing LED pattern

placed with any other 8051-compatible microcontroller. Assembly source program for microcontroller-based light chaser is generated with the help of 'Chaser designer.'

Chaser designer software is written in C++ but uses Qt library. So it can be compiled in QtCreator SDK from the link <http://qt.nokia.com/downloads>. It is a free and Open Source software. Install it

and open the file in SourceCode option with '.pro' extension, or simply double click the '.pro' file. Edit the chaserdesigner.cpp code and click 'Build' button to compile the file. The software will work on both Linux and Windows XP. In Windows XP, you don't need to compile. Just click executable file included in 'ChaserDesigner.zip' file available on www.efymag.com website to run the software.

Using the chaser designer software, LED patterns with proper delay can be easily designed. It supports up to eight channels. In a few mouse-clicks, the program generates a source code in assembly language with extension '.asm' for 8051 microcontroller. Assemble the

source code with your favorite 8051 assembler (Cross Assembler 'ASM51') to generate 'hex' file, which is then burned into the AT89551 microcontroller. Put the programmed microcontroller into assembled PCB and then power the circuit. The LEDs connected to Port 1 of AT89551 will glow in a similar pattern as the chaser running the LED pattern on PC screen. The circuit can also be connected to higher loads using triac BT136 as shown in Fig. 1.

Generation of assembly source program

On running the 'Chaser Designer.exe' application, a window appears (refer Fig. 6). On this screenshot you can see four menu bars: Designer, Organiser, Simulator and Builder. Function

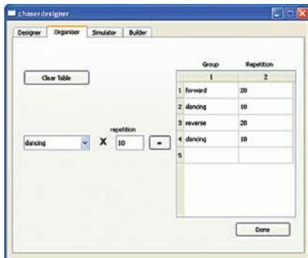


Fig. 10: Organiser window

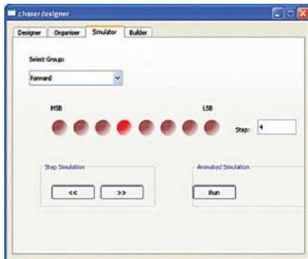


Fig. 11: Simulator window